

AI Video Analyzer - Project Notes

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Project Overview

AI Video Analyzer is an AI-powered application built using Python and Streamlit that automates video transcription, AI analysis, and report generation using Whisper, FFmpeg, and multiple AI providers.

Objectives

Automate video understanding; Support multiple AI providers; Generate summaries and insights; Export reports; Demonstrate clean software architecture.

Technology Stack

Python, Streamlit, Whisper, FFmpeg, Ollama, OpenAI, Anthropic, Pytest, GitHub Actions.

Core Features

Video upload, audio extraction, transcription, AI analysis, summaries, key points, action items, exports, logging, and progress tracking.

Architecture

Layered architecture: Streamlit UI → Services → Provider Factory → AI Providers.

Software Engineering

OOP, SOLID Principles, Factory Pattern, Dependency Injection, Clean Architecture.

Testing

Pytest, unittest.mock, unit testing, provider testing, GitHub Actions CI.

Challenges

Multiple AI providers, long video processing, transcript management, export generation.

Future Enhancements

Authentication, Docker, Cloud deployment, OCR, Speaker diarization, Batch processing.

Conclusion

The project demonstrates practical AI integration, multimedia processing, software architecture, testing, CI/CD, and documentation.

Thank You